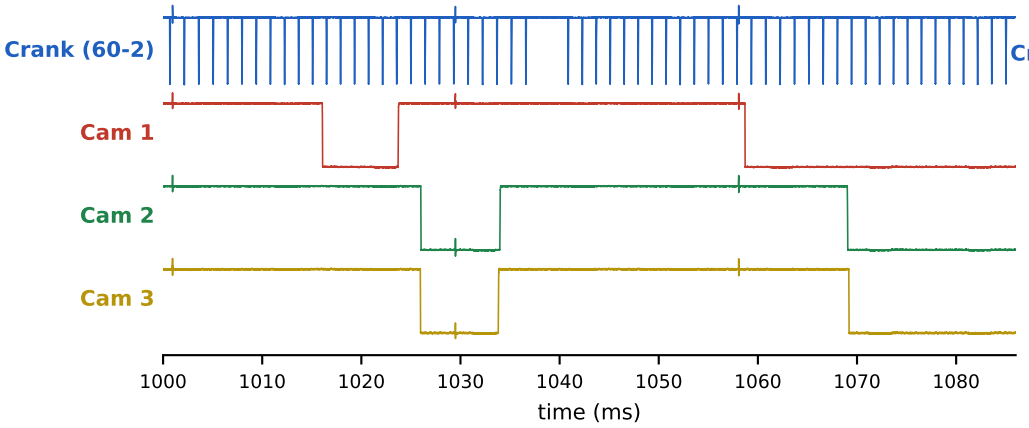
Known-good vs misfire: crank & cam scope card

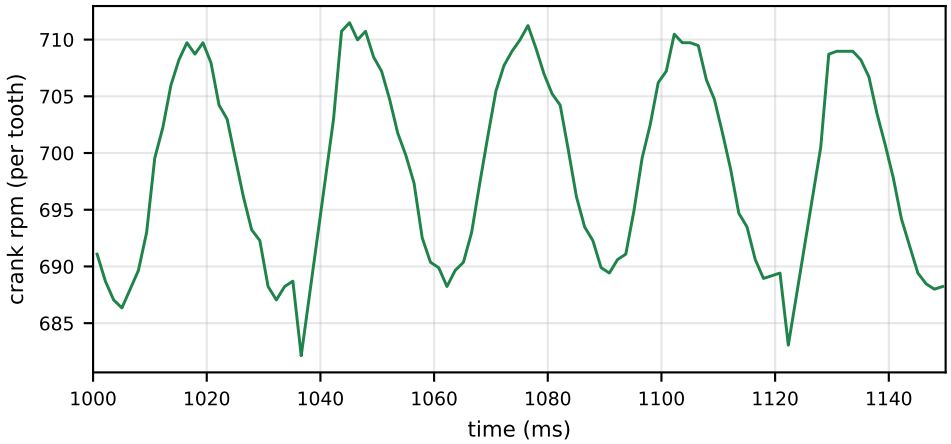
2016 Audi A6 V6 | Ch A crank (Hall, 60-2 wheel), Ch B/C/D camshaft sensors | same car, same hookup, two captures

KNOWN-GOOD (hot idle, ~700 rpm)

Raw signals: one crank revolution

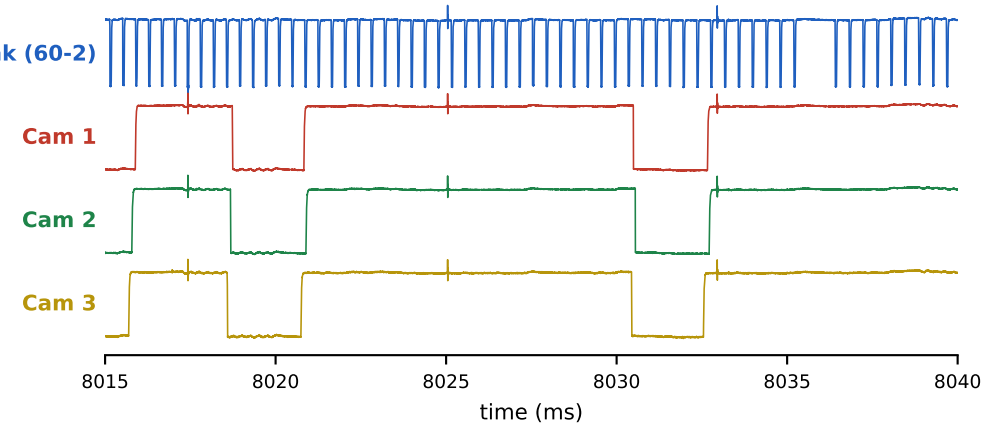


Crank speed: even pulses

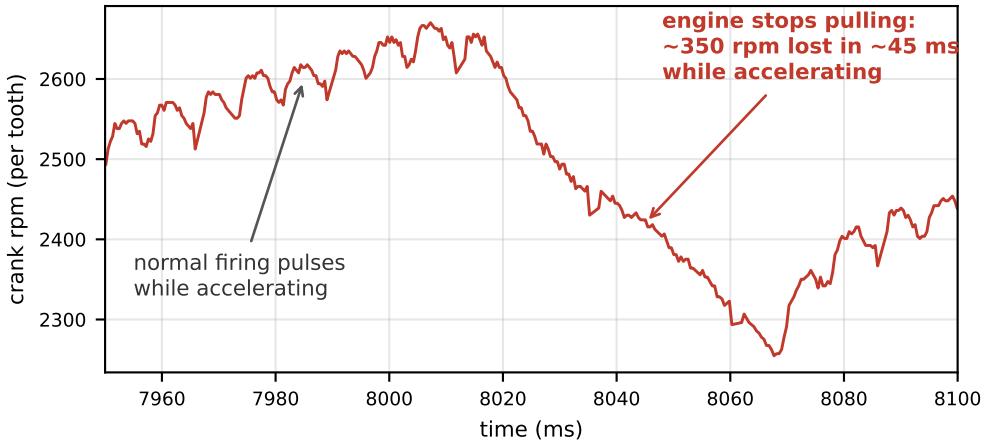


MISFIRE (accelerating, ~2,600 rpm)

Raw signals during the stumble



Crank speed: power pulses drop out



1 Read the crank, not just the code.
Each 120° of crank rotation is one V6 power stroke. Tooth-period rpm shows each firing as a small speed bump: 3 per revolution, even height at a healthy idle (±15 rpm).

2 Misfire = missing pushes.
Under load the bumps flatten and the crank decelerates in the middle of an acceleration. That is lost combustion (misfire), seen directly on the crank signal.

3 Sensors are not the problem here.
Crank and cam signals stay clean, full 0-5 V and in sync during the stumble. Move on to ignition, fuel and mechanical checks rather than replacing sensors.

Capture courtesy of Dan, Urban Auto Care. Used with permission.

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